



AI in Sports

Can it be a Game Changer?

Case Study

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Case Study

*"AI can contribute to ease certain pain points in world of sports...I think AI is also going to have a massive impact in sports in the course of the next few years."*¹

– Abhinav Bindra²

Artificial Intelligence (AI) was dramatically transforming the way we consume and analyse sports. Since its introduction, AI was growing at faster pace. It was creating a significant impact in the technical, professional, and even the personal sphere across the globe. Also, the dynamic influence of AI was felt in the sports world.³ In 2022, FIFA Virtual Assistant Referee (VAR) was used in the FIFA world cup, which helped to reach the right decision.⁴ In Wimbledon 2023, use of AI powered commentary raised the bar for sports innovation. Responding to this, IBM, collaborator for AI generated coverage in the Wimbledon 2023, indicated, "This new insight will help tennis fans to uncover anomalies and potential surprises in the singles draw, which would not be apparent by looking only at the players' ranking."⁵ Esports was considered as a medal event for the first time in the Asian Games 2023, which was held in Hangzhou, the People's Republic of China and it flaunted the advancement of digital transformation in sports.⁶

Notably, AI was becoming a game changer in sports. It helped to track the players health, diet, sleep patterns, etc., which played a key role in the performance of the players.⁷ Deepak Malik⁸ (Malik), co-founder of Stupa Sports Analytics⁹ (Stupa) further shared his experience and

¹ Sahni Jaspreet, "AI can contribute to ease certain pain points in world of sports: Abhinav Bindra", <https://timesofindia.indiatimes.com/sports/more-sports/others/ai-can-contribute-to-ease-certain-pain-points-in-world-of-sports-says-abhinav-bindra/articleshow/106885116.cms>, January 16th 2024

² He won the first individual gold medal for India in 2008, Olympics held in Beijing. He clinched the gold medal in 10-meter Air Rifle event.

³ "AI can contribute to ease certain pain points in world of sports: Abhinav Bindra", op.cit.

⁴ "Interesting Ways the AI Is Affecting the 2022 FIFA World Cup Sports", <https://www.analyticsinsight.net/interesting-ways-the-ai-is-affecting-the-2022-fifa-world-cup-sports/>, December 13th 2022

⁵ Milmo Dan and Hern Alex, "Wimbledon to introduce AI-powered commentary to coverage this year", <https://www.theguardian.com/sport/2023/jun/21/wimbledon-introduce-ai-powered-commentary-to-coverage-this-year>, June 21st 2023

⁶ Jackson Amber, "Asian Games 2023: Developing digital sports technology", <https://technologymagazine.com/digital-transformation/asian-games-2023-developing-digital-sports-technology>, September 22nd 2023

⁷ "How is AI Powering the Future of Sports?", <https://www.analyticsvidhya.com/blog/2023/05/how-is-ai-powering-the-future-of-sports/>, September 14th 2023

⁸ He was also a tennis player, coach, and a national level medallist in India.

⁹ Stupa Sports Analytics was a global sports analytics company, which was driven by love for Table Tennis.

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highlighted the need for analytics in sports. He faced difficulties like recording a match, watching it many times, etc., while communicating various instruction to a group of players. Malik also highlighted, “The entire process was exhausting. With a group of 10-15 players, it’s almost impossible to work in this manner. And there was no way to present my feedback to the players where they could understand it better.” However, with the help of AI tacking software, it became easier to communicate various data points such as movement of the players, angle of elbow, etc. from a recorded match to the players as well as coaches. While highlighting the landmark role of AI in sports, Malik said, “A player may not necessarily agree with a coach’s suggestions at all times. But now there’s data and videos to back it up, all available at a click. Besides being efficient, it has also changed the way the game is approached.”¹⁰

But, privacy issues, biasness, standardisation issues etc., were some of the key challenges with the use of AI in sports.¹¹ Responding to this, Ernesto Luna Madrid, Sports Lead at Globant¹² pointed out, “The rapid advancement of AI surprises those at its forefront. Predicting the future is challenging, but the potential for AI to redefine the sports industry is undeniable.”¹³

AI: Transforming the Sports Industry

The size of AI market particularly in the sports industry was estimated at \$5.93 billion in 2024 and it was anticipated to reach \$20.94 billion by 2029. According to the analysts at Modor Intelligence¹⁴, AI market in sports industry would grow at a CAGR 28.69% during 2024-29. Notably, increasing use of technology specifically in well developed sports was the key reason behind the rapid growth of the AI market in the sports segment.¹⁵ The Business Research Company¹⁶ further mentioned that North America was the leading region in the world for the use of AI in sports (in 2023) and Asia-Pacific was the fastest growing region for the use of AI in the sports industry.¹⁷ By witnessing the growth of the AI market in the sports industry, Ayesha Khanna (Khanna), Council Member, Forbes revealed, “With an ever-growing sports fanbase and constant AI advancements, it is inevitable that sports and AI will continue to intertwine, shaping existing and emerging industry trends.”¹⁸

AI Use in the Sports Industry

Technically, use of AI in sports referred to the use of AI techniques like machine learning, computer vision, etc., which helped to analyse as well as process data from number of sources

¹⁰ Desai Shail, “Future with AI | Champions of sports analytics want to change how cricket, boxing, table tennis, football are played, and viewed”, <https://www.moneycontrol.com/news/trends/features/sports-ai-sports-analytics-for-cricket-table-tennis-football-12007531.html>, January 6th 2024

¹¹ Winn Nick, “Revolutionising sports with AI: The use cases, challenges and future”, <https://www.infront.sport/blog/sports-technology/revolutionising-sports-with-ai-the-use-cases-challenges-and-future>, February 21st 2023

¹² Founded in 2003, Globant was one of the leading IT and software development companies in the world.

¹³ Wolff Evan, “The Future: How AI in Sports is Changing the Game for Athletes and Fans”, <https://stayrelevant.globant.com/en/insights/reports/the-impact-of-ai-in-the-sports-industry-is-immense/>, August 10th 2023

¹⁴ One of the leading market research firms in the world.

¹⁵ “AI Market In Sports Market Size”, <https://www.mordorintelligence.com/industry-reports/artificial-intelligence-market-in-sports/market-size>, 2023-24

¹⁶ It was one of the leading research firms in the world having offices in India, USA and the UK.

¹⁷ “AI in sports global market report 2024”, <https://www.thebusinessresearchcompany.com/report/ai-in-sports-global-market-report>, January 2024

¹⁸ Khanna Ayesha, “Can AI Score Big In The Future Of Sports? Five Key Trends Shaping The Industry”, <https://www.forbes.com/sites/forbestechcouncil/2023/09/27/can-ai-score-big-in-the-future-of-sports-five-key-trends-shaping-the-industry/?sh=420d718d440c>, September 17th 2023

using algorithms. Software, hardware, and services were the three major types of solutions offered by AI in the sports industry.¹⁹ Introduction of AI in sports had started in the middle of the 20th century. In 1950, Alan Turing, known as the father of computer science, used AI in his 'imitation game'^{20, 21, 22} Subsequently, in 1966, Joseph Weizenbaum, a computer scientist at MIT created Eliza chatbot, known as the first program, that allowed a type of possible conversation between human beings and machines.²³ (Annexure I).

In 1997, Prozone, the first computer vision player tracking system was introduced in high profile sports. Prozone initially partnered with the Premier League team Derby County²⁴. Subsequently, in 2001, Hawk-Eye computer vision technology (Hawk-Eye) was introduced officially in tennis.²⁵ However, use of Hawk-Eye was discussed especially after the 2004 US Open Women's quarterfinal when Serena Williams lost to Jennifer Capriati. Despite some disagreements, the International Tennis Federation (ITF) authorised the use of Hawk-Eye technology on tour in 2005.²⁶ And the US Tennis Open became the first grand slam to use the technology in 2006. Hawk-Eye technology became popular in sports. It was subsequently used in cricket, volleyball, and football etc.²⁷

Further, in 2007, Ari Kaplan (Kaplan), advisor at Houston Astros²⁸ used AI to support big league promotion of Hunter Pence, a former professional baseball right fielder and designated hitter. And AI was used for the first time 'in player personnel decisions'.²⁹ In 2011, WSC sports³⁰ used AI for game highlights.³¹ Regarding this, Jo Lemire, senior writer at Sports Business Journal³² mentioned, "WSC sports... ingests audio and video, recognizes the action and excitement, adds metadata tags for the archives and then generates video reels based on requested criteria, all without human intervention beyond the initial query. It has described its technology as 'automagical'."³³ Subsequently, AI powered automated camera system was introduced to cover the sports.³⁴

¹⁹ "AI in sports global market report 2024", op.cit.

²⁰ It was also known as Turing Test, which used machine in a three person game and the machine's intelligent behaviour was almost like a human being.

²¹ Dockterman Eliana, "The True Story of The Imitation Game", <https://time.com/3609585/the-true-story-of-the-imitation-game/>, November 28th 2014

²² "Timeline of AI use in sports", <https://www.sportsbusinessjournal.com/Articles/2023/12/04/ai-timeline#:~:text=2012%3A%20Spiideo%20introduces%20the%20first,camera%20system%20for%20tracking%20sports.&text=Messenger%20and%20Sapiens%20begin%20proliferating,Satisfi%20Labs'%20first%20sports%20client,> April 12th 2023

²³ Jarow Oshan, "How the first chatbot predicted the dangers of AI more than 50 years ago", <https://www.vox.com/future-perfect/23617185/ai-chatbots-eliza-chatgpt-bing-sydney-artificial-intelligence-history>, March 5th 2023

²⁴ Started in 1884, Derby County was one of the popular and best football clubs in the world.

²⁵ "Timeline of AI use in sports", op.cit.

²⁶ "The Impact of the Hawk-Eye System In Tennis", <https://trainingwithjames.wordpress.com/research-papers/the-impact-of-the-hawk-eye-system-in-tennis/>

²⁷ Chen Yuanzhou, et al., "Monocular Visual Analysis for Electronic Line Calling of Tennis Games", <https://arxiv.org/ftp/arxiv/papers/2107/2107.09255.pdf>, 2021

²⁸ Professional baseball team of USA based at Houston.

²⁹ "Timeline of AI use in sports", op.cit.

³⁰ One of the leading software development companies in the world that automated the creation as well as distribution of highlights of over 250 leagues and broadcast across the globe as of 2024.

³¹ "Timeline of AI use in sports", op.cit.

³² One of the leading journals in the world that covered business, analysis, news, networking and data related to sports industry.

³³ Lemire Joe, "A new world of artificial intelligence", <https://www.sportsbusinessjournal.com/Articles/2023/12/04/artificial-intelligence>, April 12th 2023

³⁴ "Timeline of AI use in sports", op.cit.

Zone 7, an AI powered human performance platform founded in 2017 created revolution by unwrapping its cutting-edge breakthrough technologies in the sports industry. The company 'developed and applied proprietary AI' to forecast injury risk for the players. The dynamic technology helped various stakeholders of sports including 'athletes, coaches and medical teams' and improved the performance of players. Regarding Zone 7, Tal Brown, CEO and Co-Founder of Zone7, said, "Zone7 is a proprietary AI solution that standardizes disparate datasets, analyzes them to uncover patterns invisible to the human eye, and makes proactive recommendations to improve output, detect burnout risk and mitigate injuries. We call this approach 'Predictive Guidance'." Subsequently, the company started working with number of elite teams across the 'EPL, MLB, MLS, La Liga, Serie A, NCAA, the Olympics and professional cycling'.³⁵

Further, to offer the sports lovers an exciting experience, in 2018, PGA Tour³⁶ used the technology of Narrative Science³⁷ to capture written recaps of every player.³⁸ Later on (in 2019), PGA Tour worked in collaboration with various technology platforms to capture every shot for each player. While disclosing the AI integrated plan of PGA Tour for the upcoming matches, Scott Gutterman, Vice President of Digital Operations, PGA Tour said, "A number of ways! One is exposing our fans to their favorite player no matter where they are on the course for however long they want to watch. The second thing is that there are roughly 32,000 shots hit at a PGA TOUR event, in particular, THE PLAYERS Championship. It used to be that you would only see about 20% of those shots. But what we're going to do now, along with the live streaming, is we're going to cut every shot as a highlight that will go onto PGA TOUR digital platforms and partner platforms."³⁹

In meantime, the world grappled with the COVID-19 pandemic. During the midst of the pandemic period, Cleveland Browns⁴⁰ approached wicket⁴¹ to help the team to get back the football lovers in the stadium by following the pandemic norms like social distancing, minimum physical contacts, etc. And "Wicket worked with the Browns to test and deploy a facial ticketing solution that allowed fans to just walk up, have their face authenticated (even with a face mask on), and walk into the stadium in less than a second — no showing or sharing mobile bar codes, removing face masks, or congregating in long ticketing lines," mentioned Wicket on its platform.⁴²

Notably, in 2022, ChatGPT (Chat Generative Pre-trained Transformer), a remarkable AI tool was launched by OpenAI⁴³. ChatGPT was able to follow an instruction as well as generate responses.⁴⁴ Regarding ChatGPT, Ethan Mollick, an associate professor of management, Wharton School of the University of Pennsylvania, said, "We're hitting a tipping point for artificial intelligence: With ChatGPT and other AI models that can communicate in plain English, write and revise text, and write code, the technology is suddenly becoming more

³⁵ Georgiadis Fotis, "The Future Is Now: Tal Brown of Zone7 On How Their Technological Innovation Will Shake Up The Sports Scene", <https://medium.com/authority-magazine/the-future-is-now-tal-brown-of-zone7-on-how-their-technological-innovation-will-shake-up-the-a80ed799ee1a>, February 28th 2021

³⁶ It was the organiser of professional golf tours across the United States and North America.

³⁷ Based in Chicago, USA, the Narrative Science was a leader in automated narrative generation for the enterprise.

³⁸ "Timeline of AI use in sports", op.cit.

³⁹ Hashtag Team, "How the PGA TOUR Is Planning to Share 32,000 Highlight Clips During THE PLAYERS Championship", <https://www.hashtagsports.com/read/pga-tour-players-championship-highlights>, December 3rd 2019

⁴⁰ A professional football team of the USA based in Cleveland.

⁴¹ It was one of the leading computer vision AI companies that offered face authentication technology.

⁴² "How We Got Started", <https://www.wicketsoft.com/who-we-are/>

⁴³ OpenAI was the most popular AI research and deployment company in the world.

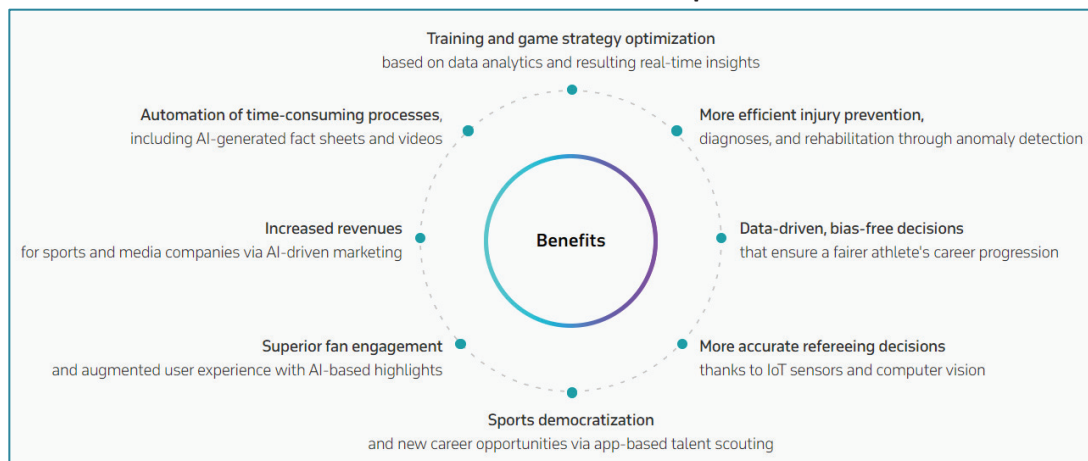
⁴⁴ "Introducing ChatGPT", <https://openai.com/blog/chatgpt>, November 30th 2022

useful to a broader population of people.”⁴⁵ In 2023, WHOOP, an American wearable technology company announced to partner with OpenAI to launch the AI powered coach known as WHOOP Coach, which would offer more personalised data insights and recommendations. Responding to this, Victoria Song, a senior reporter at *the Verge*, highlighted, “Like ChatGPT, Whoop Coach functions as a health and fitness-focused chatbot that pulls from your Whoop data. One potential use case is asking it to design custom plans, routines, and recipes depending on your goals.”⁴⁶

Factors Driving the Use of AI in Sports

Further, PwC report pointed out that AI entered into every facet of sports, starting ‘from the competition and game play to the show and entertainment business’. AI powered tools were used in various major sports disciplines such as soccer, baseball football, cricket, etc. Regarding the use of AI in sports industry, Aleksandr Ahramovich, Head, AI/ML Center of Excellence, itransition⁴⁷, mentioned, “The adoption of artificial intelligence in sports helps increase athlete performance and safety, enhance game planning, and promote fan engagement.”⁴⁸ (Exhibit I).

Exhibit I
Benefits of the Use of AI in Sports



Source: Ahramovich Alexandr, “AI in sports: top use cases, real-life examples & adoption challenges”, <https://www.itransition.com/ai/sports>, November 15th 2023

AI was widely used to analyse vast amount of data from various sources including cameras, videos, sensors, etc., which was further accessed by the coaches as well as players to improve the athlete’s performance.^{49,50} In addition to finding strength and weaknesses, the performance analysis also helped to develop competitive strategies against the rival team. By analysing the player’s movement and his biomechanical data, the players could identify

⁴⁵ Mollick Ethan, “ChatGPT Is a Tipping Point for AI”, <https://hbr.org/2022/12/chatgpt-is-a-tipping-point-for-ai>, December 14th 2022

⁴⁶ Song Victoria, “Whoop is adding a ChatGPT-powered ‘coach’”, <https://www.theverge.com/2023/9/26/23888984/whoop-coach-chatgpt-ai-fitness>, September 26th 2023

⁴⁷ One of the leading global software and engineering companies served over 800 clients across the world.

⁴⁸ Ahramovich Alexandr, “AI in sports: top use cases, real-life examples & adoption challenges”, <https://www.itransition.com/ai/sports>, November 15th 2023

⁴⁹ Goldberg Josh, “Artificial Intelligence (AI) & Its Impact on the Sports Industry”, <https://www.lexology.com/library/detail.aspx?g=404ee360-cdc8-4772-900c-eeecdca6874b9>, March 20th 2023

⁵⁰ Kumari Priyanka, “6 AI In Sports Use Cases Using Computer Vision”, <https://www.labellerr.com/blog/6-smart-sports-use-cases-using-vision-ai/>, October 10th 2023

harmful movements and protect themselves from possible injuries.⁵¹ In 2022, FIFA World Cup, AI was used to analyse the performance of the players through FIFA Players App. In addition to this, AI was used to enhance the experiences of the viewers with the help of video assistant referee, sensor embedded football and control of temperature in the stadium.⁵² **(Annexure II)**. Besides, AI helped to design personalised training programme for individual players. **(Annexure III)**. Sports lover could easily book a ticket, get live updates, and buy merchandise with the help of chatbots. In addition to this, use of AR/VR offered an immersive experience to the fans.⁵³ **(Exhibit II)**.

Exhibit II Application of AI in Sports

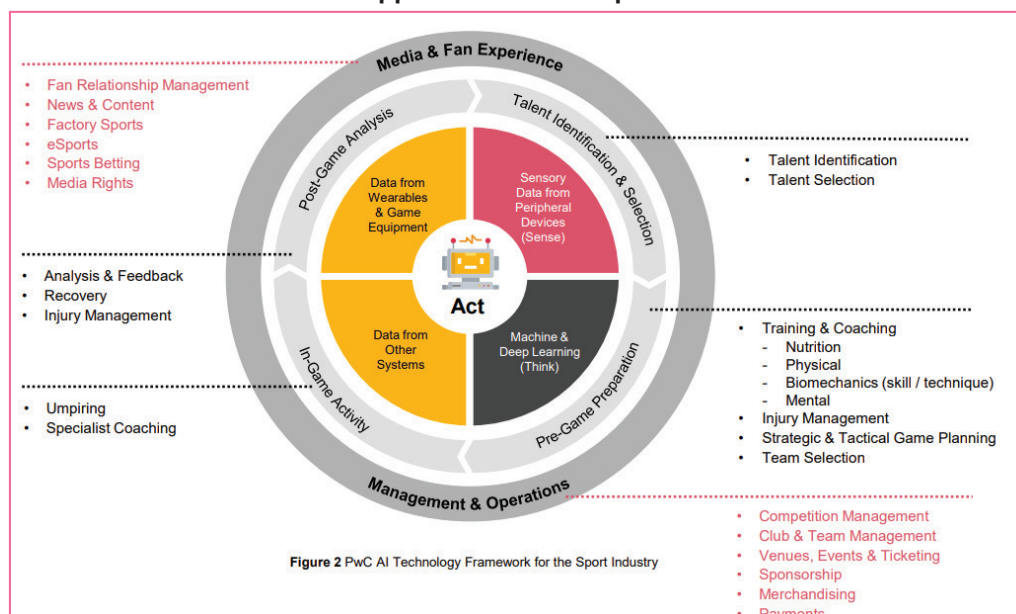


Figure 2 PwC AI Technology Framework for the Sport Industry

Source: "Artificial Intelligence Application to the Sports Industry",

<https://www.pwc.com.au/industry/sports/artificial-intelligence-application-to-the-sports-industry.pdf>, February 2019

Another remarkable contribution of AI in sports was to improve as well as augment the various equipment used in sports. Like the Jacquard tag of Google was inserted in shoe insoles of football players to monitor their movements such as kicking and running. These movements were further recorded 'into the virtual world for both better fan engagement and player insights'. In case of golf, the AI powered solutions of Altair combined 'simulation, machine learning and data analytics' to optimise club design as well as performance. The Hitëkw tennis racquet was further designed for 'text-to-image models like DALL-E and Midjourney'. Players in case of Formula 1 used AI simulations of AWS, Dell, and Oracle to optimise the 'race strategies by analysing key variables like weather, competitors, pit stops, track conditions, collisions and mechanical issues'. In addition to this, use of simulations helped the players to identify the car weaknesses as well as the potential failures. It also helped the teams to play 'within budget constraints'. While discussing various significant contribution of AI in the sports industry, Khanna, further revealed, "AI's transformative potential in sports judging continues to evolve as it focuses on providing judgment free from human error and biases."⁵⁴

⁵¹ "Artificial Intelligence (AI) & Its Impact on the Sports Industry", op.cit.

⁵² Joshi Naveen, "How FIFA Used AI at the 2022 World Cup", <https://www.allerin.com/blog/how-fifa-used-ai-at-the-2022-world-cup>, March 5th 2023

⁵³ "How is AI Powering the Future of Sports?", op.cit.

⁵⁴ "Can AI Score Big In The Future Of Sports? Five Key Trends Shaping The Industry", op.cit.

Will AI Revolutionise the Sports Industry?

Critics further revealed that the world witnessed the most controversial goal in the 1986 FIFA World Cup. Known as infamous, 'Hand of God', the goal was scored by Diego Maradona (Maradona) against England.⁵⁵ Maradona had used his hand to score the 'Goal of the Century', which was un-noticed by the referee.⁵⁶ According to Alberto Rizzoli (Rizzoli), CEO at V7⁵⁷, "Today such a goal would have been chalked off with the use of AI Referees in just a matter of minutes." Even, according to researchers, NBA officials had 'missed around or incorrectly called' around 2197 plays out of 26,822 plays (during March 2015-June 2018), which led to almost 1.49% wrong decisions, a huge in sports. Responding to this, Rizzoli mentioned, "It can be the deciding factor between a team winning or losing, especially in a game like NBA where matches often go right down to the wire. Every point matters—and every refereeing call matters." Again, during 2021, elite sports teams like Southampton FC had faced entry delays. Many people were left stuck after the game had started. And Southampton FC was compelled to refund money to the sports enthusiasts. While explaining the role of AI in addressing ticketing issues, Rizzoli highlighted, "Wicket's facial ticketing platform lets fans upload an image of their faces and the barcode number for their tickets to a game. This makes stadium entry way more efficient and prevents bottlenecks."⁵⁸

Benefits and Costs of Using AI in Sports

AI was revolutionising the way people viewed and engaged with sports.⁵⁹ Notably, in 2023, Generative AI (GenAI) startups were drawing more investment than venture capital investment in sports. In 2022, around \$783.5 million of VC funding was poured in sports AI, which was estimated at only around \$152.7 in 2023 (till mid-October), mentioned by analysts at PitchBook⁶⁰.⁶¹ (Annexure IV). Further, witnessing the remarkable transformation in the sports industry with the progress and development of AI, Kobe Bryant⁶², said, "I believe that artificial intelligence in sports will not replace human intelligence, but it will empower it, making athletes and teams smarter and more competitive."⁶³

According to analysts at International Data Corporation (IDC)⁶⁴, spending on GenAI would be estimated at around \$143 billion by 2027. GenAI helped to redefine the way contents were created and distributed in every sector including sports. With the help of GenAI, sports

⁵⁵ "Diego Maradona's ball used to score the 'Hand of God' goal in 1986 may fetch £3 mn", <https://economictimes.indiatimes.com/magazines/panache/diego-maradonas-ball-used-to-score-the-hand-of-god-goal-in-1986-may-fetch-3-mn-at-auction/articleshow/94859951.cms?from=mdr>, October 14th 2022

⁵⁶ Adhikari Somak, "Rewind To 1986 - Diego Maradona's 'Hand Of God' Stunned England At The FIFA World Cup", <https://www.indiatimes.com/sports/kick-off/30-years-to-the-day-diego-maradona-s-hand-of-god-stunned-england-at-the-fifa-world-cup-257175.html>, November 16th 2022

⁵⁷ One of the powerful AI training data platforms, which used to produce high-quality image as well as video data annotations to train computer vision models.

⁵⁸ Rizzoli Alberto, "7 Game-Changing AI Applications in the Sports Industry", <https://www.v7labs.com/blog/ai-in-sports>, October 1st 2021

⁵⁹ Nagar Tarun, "What Are The Benefits and Cost of AI in Sports Industry?", <https://devtechnosys.com/insights/cost-of-ai-in-sports-industry/>, October 30th 2023

⁶⁰ Was one of the leading private market financial platforms in the globe.

⁶¹ Fleming Margaret, "16 sports AI companies that have scored big valuations, according to new data on VC investment in the sector", <https://www.businessinsider.com/highest-valued-sports-ai-companies-globally-vc-activity-2023-10?IR=T>, October 25th 2023

⁶² Popular American professional basketball player. He died in a plane crash in 2020.

⁶³ "What Are The Benefits and Cost of AI in Sports Industry?", op.cit.

⁶⁴ Founded in 1964, ID was one of the leading firms in the world that offered services on market intelligence, advisory services, and events for the information of technology, communication technology and consumer technology markets.

broadcasting was witnessing a profound change. Regarding GenAI, Gary Warech (Warech), head of sport & entertainment at Veritone⁶⁵ highlighted, “Through GenAI, archives can be rejuvenated and showcased innovatively through curated highlight reels, nostalgic montages, or other audio and video content types. This approach revitalises past content, making it relevant for present-day audiences, and helps carve out avenues for sustained revenue generation through sponsorships or licensing opportunities.” Moreover, the versatile content created by GenAI would attract large viewers, which would increase revenue generation. Warech disclosed, “Sports broadcasting stands on the cusp of a new era, and GenAI offers a new way to approach content that’s abundant, finely tailored, deeply engaging, and ripe for monetisation.”⁶⁶ In addition to this, AI would help to enhance efficiency as well as the accuracy of the contents, which would have a positive impact on the broadcasters. Further, Luna Wolfe, writer specifically for AI at *Medium.com*⁶⁷, pointed out the key contribution of AI in sports broadcasting and said, “AI’s automation of tasks such as highlight generation and real-time analytics frees up valuable human resources... Broadcasters can allocate their time and effort to creating more engaging content, ultimately improving the quality of sports coverage.”⁶⁸ (Exhibit III).

Exhibit III
Sports Broadcasting and AI

Technologies	In Detail
Automated Highlights (MVP)	➤ Generate highlights from games automatically. ➤ Enable human editors to focus on more creative tasks. ➤ Fans would be able to access the highlights they crave and would enhance their viewing experience. ➤ Enable the broadcasters to improve their operational efficiency by saving both time and money.
Virtual reality (VR) and augmented reality (AR)	➤ Would create immersive VR and AR experiences for sports lovers. Would enable the sports enthusiasts to witness the action up close and personal with their favorite teams. ➤ Would be able to minimise the line between being in the stands and watching from home.
Real-Time Analytics	➤ Would enable the viewers with in-depth player statistics and game predictions, enriching broadcasters’ commentary, and analysis. ➤ Sports lovers could leverage this data to make informed bets, adding excitement to the game.
Intelligent Recommendations Tailored for Individual	➤ Would be able to recommend games, teams, and players based on individual interests.
Personalised Commentary	➤ Would be able to tailor commentary to personal liking for teams, individual player, or sports. ➤ This personalized touch would enhance viewing experience

Source: Wolfe Luna, “AI in Sports Broadcasting: How Technology is Changing the Way We Watch the Game”, <https://medium.com/@lunawolfe01/ai-in-sports-broadcasting-how-technology-is-changing-the-way-we-watch-the-game-ab64e4711d3a>, September 11th 2023

⁶⁵ It offered human-centered AI solutions.

⁶⁶ Warech Gary, “COMMENT: The future of sports content with generative AI”, <https://www.broadcastnow.co.uk/production/comment-the-future-of-sports-content-with-generative-ai/5187092.article>, October 20th 2023

⁶⁷ One of the popular open platforms that enabled the readers to find dynamic thinking, and enabled the expert to share their writing on any topic.

⁶⁸ Eddins Russ, “What does the future of sports broadcasting look like?”, <https://thedmonline.com/what-does-the-future-of-sports-broadcasting-look-like/>, October 8th 2023

With the help of computer vision, AI was able 'to track human motion' in sports during play. It helped to capture spatiotemporal trajectory of every player, which helped the coach to get insights about the performance of a player.⁶⁹ It could also help in grass-root level coaching. Abhinav Bindra said, "What we will see is AI's role in coaching, it will actually help democratize sport to a great degree. Talk about grassroot-level coaching, sometimes you don't have quality coaches. Imagine if AI could play a significant part there and guide young athletes in the right direction, making sure they have the right focus, right direction. It can play a very positive role there."⁷⁰

Notably, in 2020, the size of 'the player movement tracking market' was estimated at around \$3,450 million and it was anticipated to touch almost \$13,252.53 million in 2026. 'Sensor tracking and optical tracking', in hockey was tracked by the National Hockey League with the use of a hybrid system. Further, due to the distribution of data-driven information, both the player and the team would be able to predict the action of an opponent, which would help to enhance the formulation of strategy. The AR and VR market was anticipated to grow rapidly. The VR market in sports as well as entertainment in the world was anticipated to touch \$56.7 billion by 2031. Popular sports channels like ESPN and sports team like NFL and NBA, etc., were implementing AR/VR to boost fan engagement.⁷¹ In addition to this, AI helped in personnel decision. Kaplan explained the importance of AI in player selection and said, "If you have a player who has 20 reports on them over the years, it's frustrating for me to go through them... they said, 'He is good at this. He's inconsistent. But he has great bat speed. But occasionally he's lazy running the base.' I can't make heads or tails over what's the bottom line..So with gen AI, I could just summarize pros/cons or give a sentiment score. 'This one report is 10 out of 10. But over a dozen scouting reports, the consensus is he's a 6 out of 10.'"⁷²

Challenges

In spite of unleashing the potential of players, sports lovers and teams, AI in sports entangled with issues related to data privacy and security. Data used in AI were collected from various sources across the globe, which further led to misuse and data leakage. For example, in a sports streaming millions of people were able to get personalised services and cyberattack might victimise the personal data. AI in sports also raised the question of biasness and fair judgement. Regarding this, 'Analytics Vidya', mentioned, "Machines can do the human job but are still very much reliant on humans to enter data. AI systems may encourage existing biases in training if the data fails to represent diversity in athletes."⁷³ Chase Parham, writer, and editor at *Rivals.com*⁷⁴, raised questions about the accuracy of AI in broadcasting and highlighted, "AI 'just isn't good enough.' AI can track stats just fine, but when it comes to writing storylines and game recaps, it is too inaccurate for his liking."⁷⁵

Further, any product including AI used in sports for luring fans or driving business should go through a rigorous verification. While explaining this, Shaown Nandi, director of technology for industries and strategic accounts, AWS, highlighted, "The challenge with generative AI — and where you see a lot of fear and uncertainty being thrown out — is, because it's creating new

⁶⁹ "How is AI Powering the Future of Sports?", op.cit.

⁷⁰ "AI can contribute to ease certain pain points in world of sports: Abhinav Bindra", op.cit.

⁷¹ "How is AI Powering the Future of Sports?", op.cit.

⁷² "A new world of artificial intelligence", op.cit.

⁷³ "How is AI Powering the Future of Sports?", op.cit.

⁷⁴ Founded in 1998, Rival.com was a popular network of websites that offered sports media services for college sports fans.

⁷⁵ "What does the future of sports broadcasting look like?", op.cit.

data, you then have to start asking about, ‘Did it create it with the right guardrails? Did it understand what was accurate as a source and what was inaccurate?’”⁷⁶ Besides, app complexity and cost issues related to ‘software, personnel costs and hardware’, etc., might be one of the key obstacles to implement AI in sports. Further, investing in AI was a long term process. Even if a team was able to invest in AI, in many cases, it would be difficult for them to afford the additional expenditure for maintenance as well as upgradation. Logistics issues were also another obstacle for the implementation of AI in sports. Enough space was essential for the coexistence of players and machines both safely as well as securely.⁷⁷

Will AI Replace Humans?

Further, sports analysts pointed out that machines were mainly powered by algorithms, which required human instruction for proper results. Again, machines could not understand the personal emotions, like fatigue and process the data based on the programming, which would lead to inaccuracy.⁷⁸ For example, it might be difficult to replace human coach. However, AI could be an essential tool for offering supplementary support as well as guidance to humans. It would be difficult replace human touch in sports coaching.⁷⁹ Responding to this dilemma, Joel Shapiro⁸⁰ and Toby Velte⁸¹, explained, “While this new technology is definitely augmenting human decision-making, we don’t see it replacing general management teams in the near future — in sports or other businesses.” AI would augment human decisions.⁸² In addition to this, sometimes people made decision on the basis of personal experiences, which would not be the same for technology.⁸³ Further, Evan Wolff⁸⁴ highlighted, “In a world where technology evolves rapidly, sports organizations must stay on their toes. Falling behind in the AI race isn’t just a missed opportunity—it’s a threat to their future. The message is clear: AI isn’t just a player; it’s a game-changer in modern sports.”⁸⁵

⁷⁶ “A new world of artificial intelligence”, *op.cit.*

⁷⁷ Paulson Ole, “Will Athletes Ever Be Replaced By AI?”, <https://aiwhim.com/will-athletes-ever-be-replaced-by-ai/>

⁷⁸ “Will Athletes Ever Be Replaced By AI?”, *op.cit.*

⁷⁹ “Can AI Replace a Human Coach?”, <https://www.linkedin.com/pulse/can-ai-replace-human-coach-commitment-human-resources/>, October 11th 2023

⁸⁰ Was a clinical associate professor at the Kellogg School of Management, Northwestern University and academic director of Executive Education.

⁸¹ Founder and Chairman, Probit AI.

⁸² Shapiro Joel and Velte Toby, “Will AI Replace the Front Office in Pro Sports?”, <https://hbr.org/2023/06/will-ai-replace-the-front-office-in-pro-sports>, June 16th 2023

⁸³ “Will Athletes Ever Be Replaced By AI?”, *op.cit.*

⁸⁴ Global digital marketing and content leader, Globant.

⁸⁵ “The Future: How AI in Sports is Changing the Game for Athletes and Fans”, *op.cit.*

Annexure I
Use of AI in Sports

Year	In Detail
1950	Alan Turing used machine in “imitation game
1966	Eliza chatbot was developed at MIT
1997	Prozone, the first computer vision player tracking system used in elite sports
2001	Hawk-Eye computer vision tracking was used in tennis
2007	Used in player personnel decisions by Ari Kaplan for Hunter Pence’s big league promotion.
2009	SportVU’s ball and player tracking camera system was demonstrated for National Basketball Association (NBA) executives. By 2013, it equipped the entire league.
2011	WSC Sports introduced automating production of highlights.
2012	Spiideo company launched the first AI-powered, automated camera system for tracking sports.
2016	Chatbots (Satisfi Labs), Facebook Messenger and Sapiens started entering sports space. The Associated Press along with Automated Insights used Wordsmith tool for game stories on minor league baseball and high school sports.
2017	Zone7 became the first sport science company, which used AI to predict the of injury risk of the athlete
2018	Narrative Science was used by the PGA Tour to generate written recaps of every player after every round.
2019	Physimax, a motion capture app was unveiled, which worked by using only a smartphone camera.
2020	Wicket’s facial authentication test was used by the Cleveland Browns, a professional football team in the USA. In 2021, the team launched “Express Access” feature as part of its ‘Responsible Restart Plan’ during the pandemic.
2021	Semi-automated offside video-assisted review was used in trialed mode at the Arab Cup.
2022	ChatGPT was unveiled by OpenAI.
2023	AI-generated video commentary was introduced for events like the Masters, Wimbledon, and the U.S. Open by IBM. Whoop ⁸⁶ along with OpenAI planned to launch Whoop Coach, known as the first generative AI training assistant

Source: “Timeline of AI use in sports”,
<https://www.sportsbusinessjournal.com/Articles/2023/12/04/ai-timeline#:~:text=2012%3A%20Spiideo%20introduces%20the%20first,camera%20system%20for%20tracking%20sports.&text=Messenger%20and%20Sapiens%20begin%20proliferating,Satisfi%20Labs'%20first%20sports%20client.,> April 12th 2023

⁸⁶ Based in Boston Whoop was one of the leading wearable technology companies in the USA.

Annexure II

Uses of AI in FIFA World Cup, 2022



Source: "How FIFA Used AI at the 2022 World Cup", <https://www.allerin.com/blog/how-fifa-used-ai-at-the-2022-world-cup>, March 5th 2023

Annexure III

AI Personalised Athlete Training Across Various Sports

- The NBA along with Second Spectrum used AI-powered data to evaluate both player and team performance, empowering coaches to make real-time strategic decisions.
- Second, Spectrum also partners with the Premier League and Major League Soccer (MLS).
- HomeCourt, also NBA's official partner used AI to help basketball players to refine their shooting technique as well as to monitor their performance.
- Seattle Sport Sciences employs evaluated soccer players' skills, including foot preference, measuring first touch and mapping 23 touch points while collecting player data with the help of AI. It also partners with FIFA to calibrate goal-line technology.
- In golf, Xonic Golf's app analysed golfers' swings through computer vision AI algorithms, offering personalised PGA Pro tips based on biomechanics, akin to having an expert caddie by one's side.

Source: Khanna Ayesha, "Can AI Score Big In The Future Of Sports? Five Key Trends Shaping The Industry", <https://www.forbes.com/sites/forbestechcouncil/2023/09/27/can-ai-score-big-in-the-future-of-sports-five-key-trends-shaping-the-industry/?sh=420d718d440c>, September 17th 2023

Annexure IV**Some of the Leading Sports AI Companies in the World, 2023 (based on Valuation)**

Name of the Companies	Valuations	Country
SigmaStar	\$4.55 billion	China
SmartNews	\$1.99 billion	Japan
DeepGlint	\$1.15 billion	Beijing
Yi Technology	\$892 million	China
WSC Sports	\$695 million	Israel
Aimlabs	\$550 million	New York
Kakao VX	\$532 million	South Korea
Pixellot	\$500 million	Israel
Hudl	\$460 million	Lincoln, Nebraska
STATSports	\$320 million	Northern Ireland
Second Spectrum	\$200 million	Los Angeles

Source: Fleming Margaret, “16 sports AI companies that have scored big valuations, according to new data on VC investment in the sector”, <https://www.businessinsider.com/highest-valued-sports-ai-companies-globally-vc-activity-2023-10?IR=T>, October 25th 2023